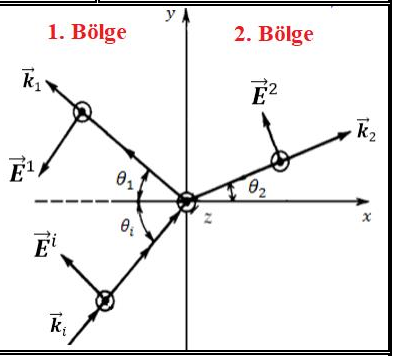
GTÜ EEM Bölümü, EMDT, Ödev 2

No, İsim- Soyadı:	210102002808, Fouad Aladhami	Tarih:	

Yanda gösterilen şekil kapsamında, 1. bölgenin elektromagnetik parametreleri $\varepsilon_1, \mu_0, \sigma_1 = 0$ ve ikinci 1. bölgenin elektromagnetik parametreleri $\varepsilon_2, \mu_0, \sigma_2 \neq 0$ olmak üzere verilmiştir. Buna göre sadece 2. bölgenin kayıplı olduğuna dikkate alarak, yansıma katsayısının sıfır olduğu ($\Gamma = 0$), Brewster açısı (θ_B) formülünü literatürden araştırarak bulunuz (Kullandığınız kitap, makale, rapor vb. kaynak bilgisini yazılmak zorundadır).

Önemli Not: θ_B açısının kompleks sayı olması halinde, gerçek açı ile ilişkisini yorumlayınız.



(A note on the Brewster Angle in lossy Dielectric Media, Ian McMichael)

For media with $\mu_{rt} = \mu_{ri}$, the Brewster angle is commonly expressed as

$$\theta_B = \tan^{-1}\sqrt{\frac{\varepsilon_{rt}}{\varepsilon_{ri}}} \quad (1)$$

r subscript: relative constitutive parameter
i subscript: the parameter of the medium in which the incident wave is propagating.
t subscript: the parameter of the medium into which the wave is transmitted.

Expressing the complex permittivity for the transmitted wave medium

$$\tilde{\varepsilon}_{rt} = \varepsilon'_{rt} - j\,\varepsilon''_{rt} \quad (2)$$



Using 1

$$\theta_B = \tan^{-1}\sqrt{\frac{\varepsilon'_{rt} - j\varepsilon''_{rt}}{\varepsilon_{ri}}} = \tan^{-1}\sqrt{\frac{|\tilde{\varepsilon}_{rt}|\exp(j\phi_{\varepsilon t})}{\varepsilon_{ri}}} = \tan^{-1}\left(\sqrt{\frac{|\tilde{\varepsilon}_{rt}|}{\varepsilon_{ri}}}\exp\left(j\,\frac{1}{2}(\phi_{\varepsilon t})\right)\right)$$

$$\theta_B = \tan^{-1}\left[\underbrace{\sqrt{\frac{|\tilde{\varepsilon}_{rt}|}{\varepsilon_{ri}}}\cos\left(\frac{1}{2}\phi_{\varepsilon t}\right)}_{a} + j\underbrace{\sqrt{\frac{|\tilde{\varepsilon}_{rt}|}{\varepsilon_{it}}}\sin\left(\frac{1}{2}\left[\phi_{\varepsilon t} - \phi_{\varepsilon i}\right]\right)}_{b}\right]$$

(3) $\theta_B = \tan^{-1}(a+jb) = \frac{j}{2}\ln\left[\frac{1-j(a+jb)}{1+j(a+jb)}\right]$, (not a real izable angle)!!

* It can see from (6) that the Brewster angle for a complex valued permettivity is a complex vake, since complex valued physical angles do not exist we can conclude that the reflection coefficient cannot truly go to zero in a soil with complex valued permettivity.
In other words, in a lossy medium there is no angle for which all of the incident TM wave power will be transmitted into that medium.

Snell law (Solution for our case) (Electromagnetic Wave theory, Dr. Serkan aksoy)

$\frac{k_2}{k_1} = \frac{\sin\theta_i}{\sin\theta_2}$, $\theta_i + \theta_2 = \frac{\pi}{2} \Rightarrow \theta_i = \frac{\pi}{2} - \theta_2 \Rightarrow r = 0$ means no reflection

$\theta_i + \theta_2 = \frac{\pi}{2} \Rightarrow \sin\theta_i = \cos\theta_2 \Rightarrow \sin^2\theta_i = \frac{1 - \sin^2\theta_2}{\cos^2\theta_2} = 1 - \left(\frac{k_1}{k_2}\right)^2\sin^2\theta_i$

$\Rightarrow \sin^2\theta_i = \frac{1}{1+\left(\frac{k_1}{k_2}\right)^2} \Rightarrow \cos^2\theta_i = 1 - \frac{1}{1+\left(\frac{k_1}{k_2}\right)^2} = \frac{\left(\frac{k_1}{k_2}\right)^2}{1+\left(\frac{k_1}{k_2}\right)^2} \Rightarrow \theta_i = \theta_B = \arctan\left(\sqrt{\frac{k_2}{k_1}}\right)$

$k_2 = \sqrt{\omega^2\varepsilon_2\mu_2 + i\omega\sigma_2\mu_2}$ (↑ lossly medium) and $k_1 = \omega\sqrt{\varepsilon_1\mu_1} = 0$ (because $\varepsilon_1, \mu_1 = 0$) (↑ lossless medium)

so $\theta_B = \arctan\left(\frac{k_2}{k_1}\right) = \arctan(\infty) = \frac{\pi}{2}$